

8. A permutation of length  $N$  is a *one-to-one onto* mapping  $\pi$  from  $0, 1, 2, \dots, N-1$  on to itself. A permutation can be applied to an array  $A$  of  $N$  integers.  $\Pi(A[i]) = A[\pi(i)]$ . If you are allowed any amount of additional storage, how can you obtain the permutation of  $A$  given  $\pi$ ? If you are allowed to use only a constant amount of additional storage, how would your solution change?
9. We all know that the standard memory allocator function 'malloc' is used for allocating memory in C programs. The size of the memory allocated in bytes is passed to 'malloc' as a parameter. What would be the program behaviour if I pass zero as the parameter to 'malloc'?
10. We are all familiar with signals such as SIGSEGV/SIGBUS, etc. I am a malicious user and I want to kill the programs of all users other than myself on the machine running Linux. Can I send a signal which will kill all other users' processes?
11. Can one use a floating-point constant value as an enumeration constant in C programs?
12. Remember that functions allocate storage on the stack to hold the activation records. You are asked to write a function which will return its stack size. How can you do this?
13. Here is a small code snippet:

```
int* foo(void* p)
{
    p = p + 3;
    return (int*)p;
}
```

Can you identify any bug in the above code?

14. Let  $A$  be a sorted array of integers. You are given an integer  $K$ . Write a program to determine whether there are two indices  $i$  and  $j$ , such that  $A[i] + A[j] = K$ . Note that  $i$  and  $j$  need not be distinct.
15. Given two strings  $s$  and  $t$ , write a

program to find all occurrences of  $t$  in  $s$ . You are originally told that  $s$  contains a maximum of 1000 characters. Now, after having written your program to solve the problem, the examiner tells you that he was wrong, and  $s$  can contain 100000 characters. Do you need to re-write your algorithm to solve the problem?

16. You are given two strings,  $s$  and  $t$ . You need to determine whether  $t$  is a cyclic rotation of string  $s$ . For instance, string  $t$  is obtained by rotating each character of string  $s$  by  $k$  positions. For example, the string 'kite' is a cyclic rotation of string 'teki'. You are told that  $N$  is the maximum size of the string. Can you write a code to determine this in  $O(N)$  time with constant additional storage?
17. You are given a list of points in a two-dimensional Cartesian plane  $\langle X1, Y1 \rangle, \langle X2, Y2 \rangle, \dots, \langle Xn, Yn \rangle$ . You are asked to find the pair of points closest to each other in the Cartesian plane.
18. We typically use 'malloc' and 'free' to allocate/deallocate memory on the heap in C programs. However, in your case, you have been told that the dynamic memory has to be shared between two processes. Therefore, you need to do dynamic memory allocation on a shared memory segment, which can be shared between two processes. Can you write skeleton code to implement a dynamic memory allocator on a shared memory segment?
19. You are given a number 'N' and asked to write a function to determine whether it is prime or not. Now that you have written the solution, you have been told to find that 10,000th prime number.

Would it be more efficient to re-use your previous solution to solve this problem or would you create a new program to determine the 10,000th prime number? What is the runtime of your solution in either case?

20. You are given a stream of integers which come in as input. At any point of the input stream, you can be asked to determine the current maximum and the second largest number seen so far in the stream. Can you write a function to do this?

## My 'must-read book' for this month

This month's 'must-read book' suggestion comes from one of our readers, Unnikrishnan, who recommends 'Introduction to Algorithms: A Creative Approach' by Udi Manber. This book teaches you to examine each step in the algorithm design process and even takes you through the process of algorithm design by carefully crafted examples. Thank you, Unnikrishnan! I have not yet read the book but am hoping to do so in the coming months.

If you have a favourite programming book or article that you think is a must-read for every programmer, please do send me a note with the book's name, and a short write-up on why you think it is useful so that I can mention it in the column. This would help many readers who want to improve their coding skills.

If you have any favourite programming puzzles that you would like to discuss on this forum, please send them to me, along with your solutions and feedback, at [sandyasm\\_AT\\_yahoo\\_DOT\\_com](mailto:sandyasm_AT_yahoo_DOT_com). Till we meet again next month, happy programming, and here's wishing you the very best! 

## By: Sandya Mannarswamy

The author is an expert in systems software and works at Hewlett-Packard India. Her interests include compilers, multicore technologies and software development tools. If you are preparing for systems software interviews, you may find it useful to visit Sandya's LinkedIn group 'Computer Science Interview Training India' at <http://www.linkedin.com/groups?home=&gid=2339182>.